Biochemical Engineering Project Due on April 10

Design the experiments to express the following protein at the production scale. When you select an expression system, you need to consider the cost, yield, and feasibility (size, glycosylation and disulfide bond) of using the selected expression system. Your report should include

1. The (putative) function of the protein
2. DNA sequence of the protein:
   a. adjust the codons to the selected expression host.
   b. include the enzyme cutting site and other component for expression in the sequence
3. Molecular cloning procedures
   a. describe the key steps for the cloning, selection, and verification (gene and protein)
   b. provide the detailed information about expression vector and cloning site
   c. provide the sequences of both cloning primer and sequencing primer
4. Host cells and expression conditions

MNLQAQPKAQNKRKRCLFGGQEPAPKEQPPPLQPPQQSIRVKEEQYLGHEGPGGAVSTS
QPVELPPPSSLALLNSVVYGPERTSAAMLSQQVASVKWPNSVMAPGRGPERGGGGGVS
DSSWQQQPGQPPPHSTWNCHSLSLYSATKGSPHPGVGVPTYYNHPEALKREKAGGPQL
DRYVRPMMPQKVQLEVGRPQAPLNSFHAAKKPPNQSLPLQPFQLAFGHQVNRQVFRQG
PPPPNPVAAFPPQKQQQQQQPQQQQQQQAALPQMPLFENFYSMPQQPSQQPQDFGLQ
PAGPLGQSHLAHHSMAPYPFPPNPDMNPELRKALLQDSAPQPALPQVQIPFPRRSRRLSK
EGILPPSALDGAGTQPGQEATGNLFLHHWPLQQPPPGSLGQPHPEALGFPLELRESQLLP
DGERLAPNGREREAPAMGSEEGMRAVSTGDCGQVLRGGVIQSTRRRRRASQEANLLTL
AQKAVELASLQNAKDGSGSEEKRKSVLASTTKCGVEFSEPSLATKRAREDSGMVPLIIPV
SVPVRTVDPTEAAQAGGLDEDGKGPEQNPAEHKPSVIVTRRRSTRIPGTDAQAQAEDM
NVKLEGEPSVRKPKQRPRPEPLIIPTKAGTFIAPPVYSNITPYQSHLRSPVRLADHPSERSF
ELPPYTPPPILSPVREGSGLYFNAIISTSTIPAPPPITPKSAHRTLLRTNSAEVTPPVLSVMGE
ATPVSIEPRINVGSRFQAEIPLMRDRALAAADPHKADLVWQPWEDLESSREKQRQVEDL
LTAACSSIFPGAGTNQELALHCLHESRGDILETLNKLLLKKPLRPHNHPLATYHYTGSDQ
WKMAERKLFNKGIAIYKKDFFLVQKLIQTKTVAQCVEFYYTYKKQVKIGRNGTLTFGD
VDTSDEKSAQEEVEVDIKTSQKFPRVPLPRRESPSEERLEPKREVKEPRKEGEEEVPEIQE
KEEQEEGRERSRRAAAVKATQTLQANESASDILILRSHESNAPGSAGGQASEKPREGTG
KSRRALPFSEKKKKTETFSKTQNQENTFPCKKCGR